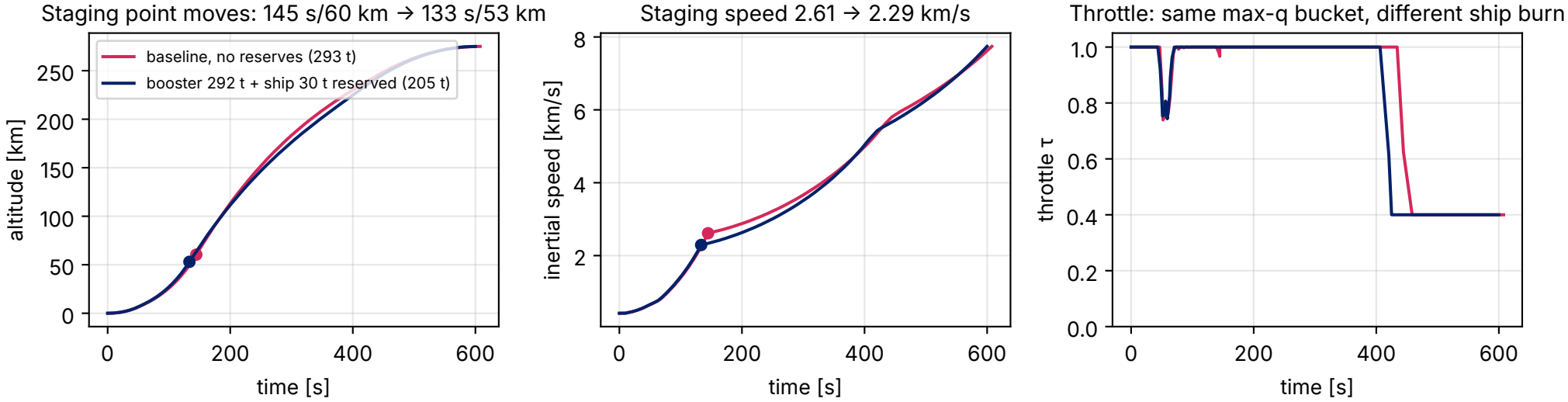
Effect of return-propellant reserves on the optimal ascent (markers = staging)
Staging point moves: 145 s/60 km → 133 s/53 km
baseline, no reserves (293 t)
booster 292 t + ship 30 t reserved (205 t)
altitude [km]
time [s]
Staging speed 2.61 → 2.29 km/s
inertial speed [km/s]
time [s]
Throttle: same max-q bucket, different ship burn
throttle τ
time [s]